

Language Models Do Not Embed Numbers Continuously

Alex O. Davies^{1,2}, Roussel Nzoyem^{1,2}, Nirav Ajmeri², Telmo M. Silva Filho²

¹Primary Author

²University of Bristol, UK

alexander.davies@bristol.ac.uk, rd.nzoyemngueguin@bristol.ac.uk

Abstract

Recent research has extensively studied how large language models manipulate integers in specific arithmetic tasks, and on a more fundamental level, how they represent numeric values. These previous works have found that language model embeddings can be used to reconstruct the original values, however, they do not evaluate whether language models actually model continuous values *as* continuous. Using expected properties of the embedding space, including linear reconstruction and principal component analysis, we show that language models not only represent numeric spaces as non-continuous but also introduce significant noise. Using models from three major providers (OpenAI, Google Gemini and Voyage AI), we show that while reconstruction is possible with high fidelity ($R^2 \geq 0.95$), principal components only explain a minor share of variation within the embedding space. This indicates that many components within the embedding space are orthogonal to the simple numeric input space. Further, both linear reconstruction and explained variance suffer with increasing decimal precision, despite the ordinal nature of the input space being fundamentally unchanged. The findings of this work therefore have implications for the many areas where embedding models are used, in-particular where high numerical precision, large magnitudes or mixed-sign values are common.

Introduction

Large Language Models (LLMs), trained on next token prediction over internet-wide data, demonstrate extraordinary emergent abilities to manipulate numbers and perform arithmetic operations beyond their training date. For this reason, they are increasingly deployed in complex safety-critical scenarios requiring complex mathematical reasoning, such as accounting (Yoo 2024), medical calculations (Khandekar et al. 2024), radiotherapy planning (Wang et al. 2025), to name but a few.

These deployments pose serious safety concerns, emphasising the need to investigate how LLMs represent numbers. One problem which plagues both the expressivity and computational efficiency of LLMs is the long-range dependency (Vaswani et al. 2017; Gu and Dao 2023). It is commonly understood that LLMs perform better with shorter prompts, or focus on a specific parts of the prompt when it is too long

(Hengle et al. 2024). That said, when inserting decimal numbers in LLM prompts, users tend to include arbitrary number of decimal places, thereby prolonging the size of length of the prompt. Knowing how well LLMs represent numbers based on their precision stands to empower both users and practitioners.

The same models are also applied in scientific domains. In some works embedding models are used as-is; Peikos, Kasela, and Pasi (2024) and Amugongo et al. (2025) use multiple models for medical document retrieval, both applications that require a complex understanding of numerical values. Other works fine-tune over domain-specific data before using a model’s embeddings, such as Lin et al. (2024) for geoscience applications and Choudhary (2024) for material property prediction and retrieval. Again, these works assume that LLMs are capable of usefully understanding continuous numerical values, despite significant differences in properties or semantic meaning across a range of numerical values. We provide examples from different scientific fields, with visualisations in Figure 1:

Example 1 (Climate Science). Atmospheric aerosol concentrations can range from 10^{-12}kg m^{-3} in exceptionally clean air (e.g. the Arctic) to 10^{-3}kg m^{-3} in urban areas. A scientist querying for “*Black carbon concentrations around $2.847 \times 10^{-9}\text{kg m}^{-3}$* ” might receive “ $2.847 \times 10^{-6}\text{kg m}^{-3}$ ” ranked as highly similar simply because both strings share the mantissa 2.847.

Example 2 (Drug Discovery). Inhibitor potencies span many orders of magnitude: picomolar compounds (10^{-12}M) are ultra-tight binders suitable for therapeutic development, while millimolar compounds (10^{-3}M) bind so weakly they’re considered inactive. A medicinal chemist searching for “*IC50 values near $0.0234\text{ }\mu\text{M}$* ” might retrieve compounds at 2.34 nM or 2.34 μM as similar matches.

There are also examples of where users would expect a more complex – and not strictly linear – interpretation of continuous values:

Example 3 (Astronomy). Stellar velocities within a galactic disk might range from -500 km s^{-1} (stars moving toward the observer) to $+500\text{ km s}^{-1}$ (stars moving away), with precision to 0.001 km s^{-1} required to detect exoplanets via Doppler wobble. An astronomer querying for “*velocity -12.847 km s^{-1}* ” might not retrieve “ $+12.847\text{ km s}^{-1}$ ” as sim-

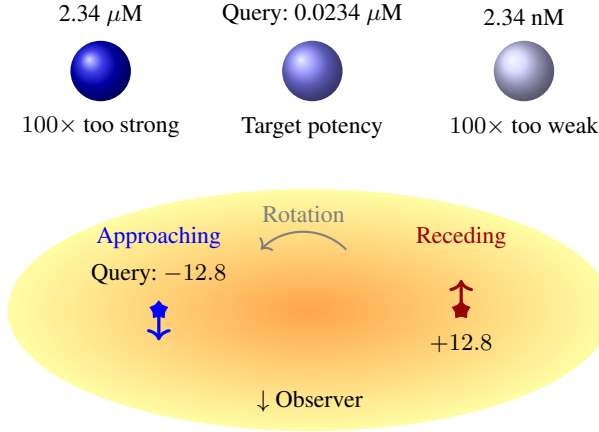


Figure 1: Visualisations of our examples for LLM embeddings in scientific knowledge applications. **Top:** In material science, the magnitude of a concentration is crucial, but repeated mantissas in numerical representations could cause incorrect retrieval. **Bottom:** In astronomy, a negatively signed value may not indicate it is semantically *opposite* to its positive counterpart, such as in measuring velocities within galactic disks.

ilar, despite the physical equivalency on opposite sides of the galactic disk, as the magnitude is opposite.

The behaviours these models exhibit in embedding numerical values is therefore crucial before application. A model which embeds strictly linearly is useful in the climate and material science examples, but would require more complex treatment where magnitude does not necessarily imply dis-similarity.

Since other recent works have highlighted representational space as critical for arithmetic (Maltoni and Ferrara 2024; Zhu, Dai, and Sui 2024), a plethora of methods have attempted to understand the semantics of the LLM embedding space. The current literature is heavily focused on the representation and processing of integers within specific arithmetic or reasoning tasks (Levy and Geva 2024; Kantamneni and Tegmark 2025). Some studies have attempted to reframe numerical representations to improve LLM performance (Schwartz et al. 2024; Zhang-Li et al. 2024), while others have brought attention to the connections to human cognition (Shah et al. 2023; AlquBoj et al. 2025). While this has yielded fascinating insights into the complex, often non-linear geometry of numerical representations, it leaves a more fundamental question unanswered: *how well do embedding models encode the basic semantic value of continuous real numbers across varying scales and precisions?*

Our work addresses this gap by providing a **general** and **lightweight** framework for evaluating the semantic fidelity of numerical embeddings. Rather than decoding a specific geometric structure (e.g., a helix or circle (Kantamneni and Tegmark 2025)), we propose a set of metrics (namely linear R^2 , PCA correlation and explained variance) that directly quantify how well an embedding captures the one-dimensional nature of a scalar value. Using these metrics we

provide a critical insight: the complex, multi-faceted representations identified in other studies (string-entanglements, periodic features) manifest as quantifiable “noise” in the embedding space. Our methodology thus offers a scalable and task-agnostic tool to measure the purity and robustness of any model’s ability to represent the simple concept of numerical magnitude.

Specifically, our contributions are as follows:

- (1) A general and lightweight framework to evaluate the fidelity of continuous embeddings, addressing a critical gap in the literature in a model-agnostic manner.
- (2) Task-independent metrics (linear R^2 , PCA correlation, and explained variance) which demonstratively quantify how well an embedding captures the ordinal, one-dimensional nature of the scalar values. These quantify the fidelity of numerical representations, enabling practitioners to better understand model limitations and optimize prompt design.
- (3) A breadth of experiment to validate our framework and the proposed metrics. We go beyond the scope of the current literature by considering positive decimals, mixed sign decimals, and mixed sign integers.

Related Work

A central challenge for Large Language Models (LLMs) is their inconsistent and often fragile ability to perform numerical reasoning. Recent research has tackled this from multiple angles: improving arithmetic performance through novel processing techniques, probing the internal geometry of numerical representations, and drawing parallels between model and human numerical cognition.

Internal Representations and Processing Methods The current theme in the literature is that LLMs introduce complex patterns into embeddings for simple 1D scalar values (Zhu, Dai, and Sui 2024). Probing experiments reveal highly complex internal structures. For instance, numbers appear to be encoded using per-digit circular representations in base 10 (Levy and Geva 2024), which helps explain why model errors are often digit-based rather than value-based. For specific operations like addition, models have been shown to develop even more intricate structures, representing numbers as a generalized helix and manipulating them with a trigonometric “Clock” algorithm (Kantamneni and Tegmark 2025). While limited to whole number representations, Kantamneni and Tegmark (2025) show that LLMs contain the ability to abstract away the continuous space. Zhou et al. (2024) identify Fourier features with varying levels of periodicity within these representations, which are later used for arithmetic operations. This internal complexity is further compounded by a fundamental ambiguity: LLM representations are often an entanglement of a number’s value and its string-like properties, where similarity is influenced by various metrics of distance (Marjeh et al. 2025).

Reformatting Numerical Representations: To improve performance in light of these complex representations, researchers have focused on reformatting inputs to align with computational logic. NumeroLogic (Schwartz et al. 2024),

for example, prefixes numbers with their digit count to provide essential place-value context upfront. In a similar vein, Little-Endian Fine-Tuning (LEFT) (Zhang-Li et al. 2024) reverses the order of digits to mimic human-like computation (least significant digit first), dramatically improving efficiency and accuracy in arithmetic tasks. Neither work, unlike ours, investigates the role of number magnitude and precision. The need for these techniques is underscored by comprehensive benchmarks which reveal that modern LLMs still fail at a wide range of basic numerical tasks. For example, the NUPA Test (Yang et al. 2024) shows broad failures beyond simple addition, while Tang et al. (2025) highlight significant error rates in the seemingly straightforward task of numerical translation, especially when dealing with large units across languages.

Experiments

Consider a real scalar number $x \in \mathbb{R}^1$, and an embedding model $f(x) \rightarrow \hat{x} \in \mathbb{R}^d$ for d the dimensionality of the model’s embedding space. Scalars x are in a set of $X = \{x_1, x_2, \dots\}$. Further, consider that $x \in X$ has a given number of integer and decimal places a and b ;

$$\begin{array}{ccc} 1234 & . & 567 \\ \text{---} & . & \text{---} \\ a = 4 & & b = 3 \end{array}$$

In this work we evaluate how accurately embedding models encode numbers x with respect to these precisions a, b and sign of the number. A diagram of our experimental framework is presented in Figure 2. We can expect that, as in prior work (Zhu, Dai, and Sui 2024; Levy and Geva 2024), a number x can be reproduced by a linear model over the embedding $\text{lin}(\hat{X}) \rightarrow X'$; perfect reconstruction is $\text{lin}(\hat{X}) = X$. More precisely, we expect that there is very good correlation between predicted values from this linear model X' and the original scalars X .

$$\text{corr}(X', X) \simeq 1 \quad (1)$$

Further, we know that there is only one component of variation, with $\text{Rank}(X) = 1$. As a result we expect that strong embeddings of numbers, with ‘understanding’ of numeric spaces, would similarly have only one component of variation, $\text{Rank}(\hat{X}) \simeq 1$. As a result, Principal Component Analysis (PCA) over the embedded scalars $\hat{X}$ should have an explained variance ratio in the first component of approximately 1:

$$\text{VR} = \frac{\lambda_0}{\sum_0^d \lambda_i} \simeq 1 \quad (2)$$

with λ_i the eigenvalues of the covariance matrix. We can also expect that the primary direction of variation in the embedding space is in-line with the original set of scalars X :

$$\text{corr}(\text{PCA}_0, X) \simeq 1. \quad (3)$$

For a perfect encoder, and continuous internal representations, we expect the relationships in Equations 1,2 and 3 to

hold true regardless of integer and decimal places a, b or the signs of the numbers.

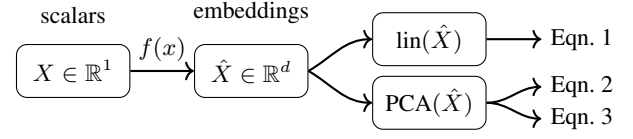


Figure 2: Framework for measuring numerical embedding quality. Scalars are embedded into high-dimensional space and evaluated using linear reconstruction and PCA to quantify preservation of numerical structure through three complementary metrics, defined in Equations (1,2,3).

Datasets We produce three datasets of scalars X . The first is positive decimals $x \in [0, 1]$ with $a = 1$ and with the precision of the decimals iteratively increased from $a = 1$ to $a = 20$. The second is mixed sign decimals $x \in [-1, 1]$ with the same precisions as the positive decimals. The third is mixed-sign scalars of varied integer and zero decimal places $b = 0, a \in [0, 20]$. The range of this dataset varies according to a , $-10^{(a)} < x < 10^{(a)}$. Across the three datasets individual scalars are sampled randomly, up to 500 samples per dataset. We adopt a 5-fold split to produce error margins.

Embedding Models In our evaluation of LLMs as embedding models for numeric data, we use models which are specifically targetted at embedding applications. We use all of the embedding models available from three public providers:

Gemini ¹Provided by Google, we use the `gemma-embedding-001` model with the default (largest) embedding dimension. Gemini embedding models are initialised from the Gemini LLM, with two rounds of contrastive training to produce the embedding model (Lee et al. 2025). Gemini embedding models are reported to out-perform competitors on benchmarks.

OpenAI ² we use the three latest models provided by OpenAI, namely `text-embedding-3-large`, a condensed version `text-embedding-3-large` and the old `text-embedding-ada-002`. No companion paper is published, but public documentation and releases suggests that OpenAI’s embedding models are based on the GPT-4 family, and are contrastively fine-tuned for embeddings (OpenAI 2025). OpenAI models are reported to out-perform competitors on benchmarks.

VoyageAI ³, provided by MongoDB, provides several text embedding models. We evaluate their non-specialist models, namely the 3.5 series (default, `lite` and `large`). No companion paper is published, with public documentation suggesting pure contrastive training for these embedding models (VoyageAI 2025). VoyageAI models are reported to out-perform competitors on benchmarks.

¹<https://ai.google.dev/gemini-api/docs/embeddings>

²<https://platform.openai.com/docs/guides/embeddings>

³<https://docs.voyageai.com/docs/embeddings>